



Service Tool Instruction

Service Tool Instruction
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Revision Level:

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Description

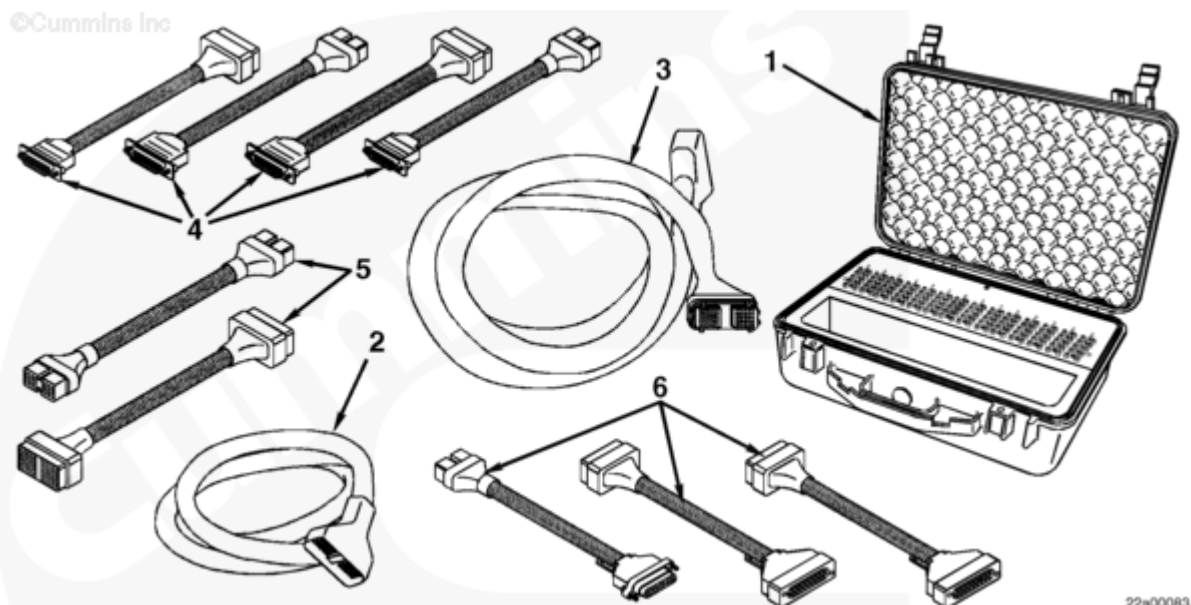
Electrical Circuit Tester Kits

Purpose

This service tool instruction provides procedures for using the electrical circuit tester kit, Part Numbers 3163594 or 3163878. The electrical circuit tester, Part Number 3163589, is portable and allows the technician to monitor voltage, current, and open circuits in the ECM harness at the ECM pins. The electrical circuit tester, Part Number 3163878, **only** allows the technician to monitor voltage in the harness and is **not** shown in the illustration below.

Note : Do not operate vehicle while tool is in use.

Both electrical circuit tester kits include an electrical circuit tester, harness adapter cable (10 feet), and ECM connector adapter cable (10 feet). The electrical circuit tester kits allow the technician to work on the following engines: ISB, ISC, ISL, ISM, ISX, Signature, QSB, QSC, QSL, and QSX. Optional adapter cables are available for power generation, Quantum™/CENSE™, and CELECT™/CELECT™ Plus.

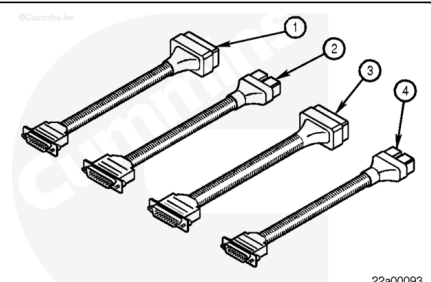


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Table 1, Electrical Circuit Tester Kit, Part Number 3163594			
Item Number	Part Number	Description	Quantity
1	3163589	Electrical circuit tester	1
2	3163590	Harness adapter cable (3048 mm [10 ft])	1
3	3163591	ECM adapter cable (3048 mm [10 ft])	1
Not shown	3163953	Storage case	1

Table 2, Items Used with Electrical Circuit Tester Kit, Purchased Separately			
Item Number	Part Number	Description	Quantity
4	3163880	Power generation adapter kit (see page 2)	4
5	3163883	Quantum™/CENSE™ adapter kit (see page 2)	2
6	3163886	CELECT™/CELECT™ Plus adapter kit (see page 2)	2
Not shown	3163878	Electrical circuit tester kit (see page 2)	1
Not shown	3164182	ISB with CM850, ISX/Signature with CM870 adapter kit	1

When using the electrical circuit tester kits for the power command control II kit, generator drive control system kit, or power generation adapter kit, it will be necessary to purchase the power generation adapter kit.

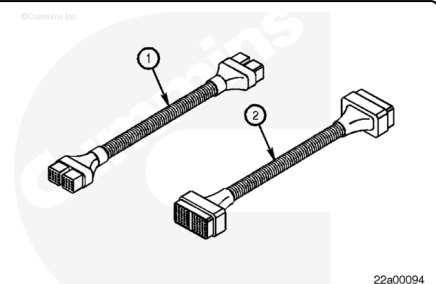


Item	Part Number	Description	Quantity
1	3163592	Harness adapter, 50-pin D-sub	1

2	3163593	ECM adapter, 50-pin D-sub	1
3	3163881	Harness adapter, 25-pin D-sub	1
4	3163882	ECM adapter, 25-pin D-sub	1

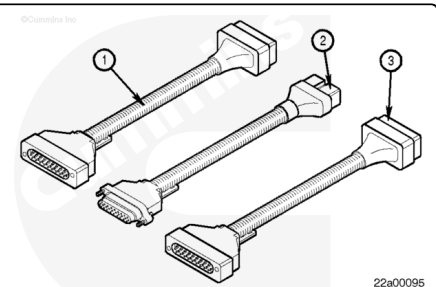
When using the electrical circuit tester kits for the QST30 generator drive or QSK19, QSK45, QSK60, and CENSE™, it will be necessary to purchase the Quantum™/CENSE™ adapter kit.

Item	Part Number	Description	Quantity
1	3163884	Harness adapter, 40-pin Deutsch connector	1
2	3163885	ECM adapter, 40-pin Deutsch connector	1



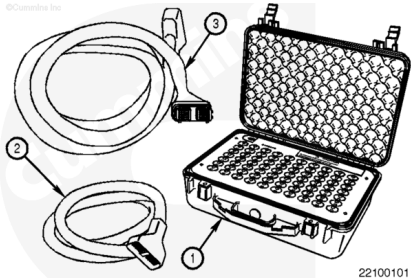
When using the electrical circuit tester kits for the CELECT™ or CELECT™ Plus system, it will be necessary to purchase the CELECT™/CELECT™ Plus adapter kit.

Item	Part Number	Description	Quantity
1	3163887	Harness adapter, 28-pin AMP connector	1
2	3163888	ECM adapter, 28-pin AMP connector	1



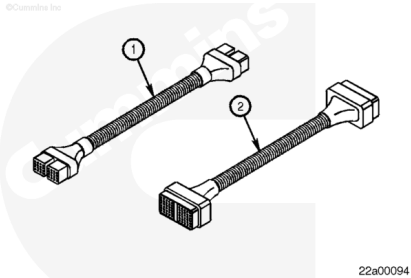
3	3163961	Harness adapter, 28-pin AMP connector	1
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This electrical circuit tester kit is used to monitor voltage **only** in the wiring harness.



Item	Part Number	Description	Quantity
1	3163879	Electrical circuit tester	1
2	3163590	Adapter cable	1
3	3163591	Adapter cable	1
Not shown	3163953	Storage case	1

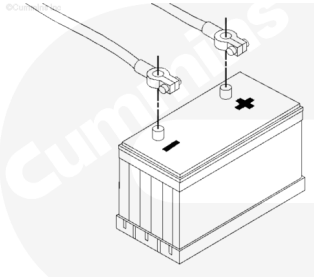
When using the electrical circuit tester kits on the 60-pin engine connector on the ISB with the CM 850 or ISX with the CM 870, it will be necessary to purchase adapter kit, Part Number 3164182.



Item	Part Number	Description	Quantity
1	3164184	Harness adapter, 60-pin Deutsch connector	1
2	3164183	ECM adapter, 60-pin Deutsch connector	1

⚠ WARNING ⚠

Batteries can emit explosive gases. To avoid personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first, and attach the negative (-) battery cable last.

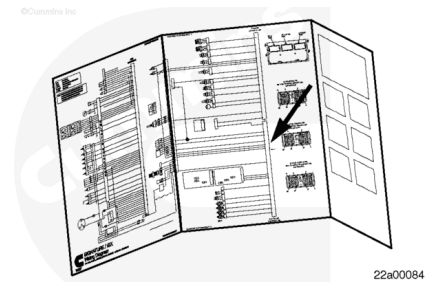


Disconnect the battery cables, negative (-) cable first.

⚠ CAUTION ⚠

To avoid severe ECM damage, refer to the appropriate wiring diagram to make certain the pin is located properly.

When disconnecting the ECM, refer to the appropriate wiring diagram to identify the correct ECM connector.

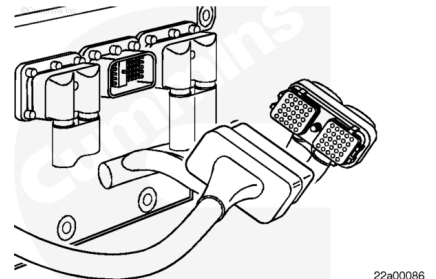


Remove the capscrew that holds the connector to the ECM.

Remove the connector.

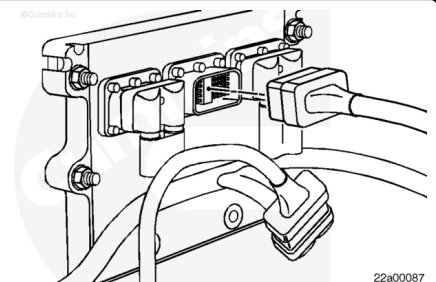
Install the connector to the harness adapter cable.

Refer to the parts list at the beginning of this instruction for the correct adapter cable part number.



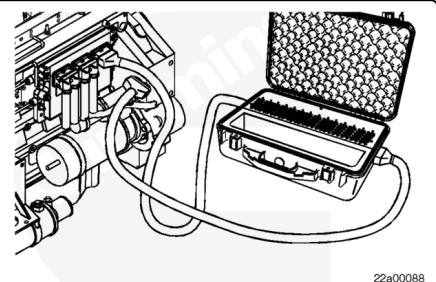
Connect the ECM adapter cable to the ECM.

Refer to the parts list at the beginning of this instruction for the correct ECM adapter cable part number.



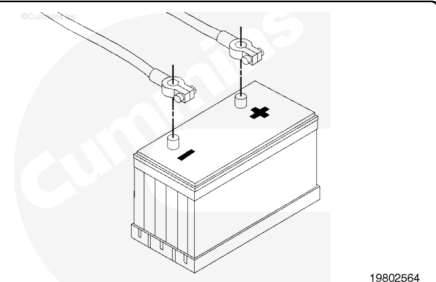
Connect the ECM adapter cable from the ECM to the electrical circuit tester.

Connect the harness adapter cable from the engine harness to the electrical circuit tester.



⚠ WARNING ⚠

Batteries can emit explosive gases. To avoid personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first, and attach the negative (-) battery cable last.



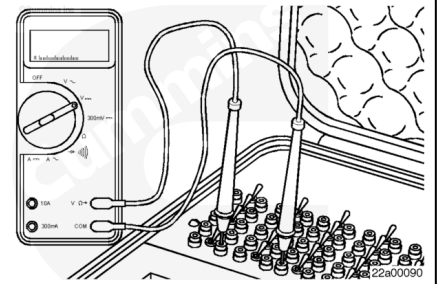
Reconnect the battery cables, negative (-) cable last.

⚠ CAUTION ⚠

To avoid severe ECM damage, refer to the appropriate wiring diagram to make certain the pin is located properly.

Use a multimeter and the electrical circuit tester, Part Number 3163589 or Part Number 3163879, to measure the voltage on a specific pin. Place the positive (+) meter lead in the desired pin. Find a ground (-) on the electrical circuit tester using the appropriate wiring diagram.

Set the toggle switch to the CLOSED position. This **only** applies to the electrical circuit tester, Part Number 3163589.



⚠ CAUTION ⚠

To avoid severe ECM damage, refer to the appropriate wiring diagram to make certain the pin is located properly.

Use a multimeter and the electrical circuit tester, Part Number 3163589, to measure the amperage on a specific pin.

Place the positive (+) meter lead in the desired pin's left test point; then place the negative (-) meter lead in the desired pin's right test point.

Set the toggle switch to the OPEN position.

